



**UNIVERSITI TEKNOLOGI MARA
FACULTY OF INFORMATION SCIENCE**

**BACHELOR OF INFORMATION SCIENCE (HONS.) INFORMATION SYSTEM
MANAGEMENT
IMS566 - ADVANCED WEB DESIGN DEVELOPMENT AND CONTENT MANAGEMENT**

**GROUP ASSIGNMENT:
SYSTEM DEVELOPMENT**

TITLE:
JOBMATCH (ONLINE JOB APPLICATION)

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1.0 Introduction

1.1 Overview of the system

JobMatch is a web-based online job portal created specifically for Universiti Teknologi MARA (UiTM) nationwide, aimed towards being a central hub for UiTM administration to promote and cope with full-time and part-time vacancies at UiTM. The system supports various employment activities in the UiTM system, from academic roles like lecturers and tutors, support and operation roles ranging from café assistants, baristas, administrative, and maintenance workers across all UiTM campuses throughout the country.

JobMatch differs from the general recruitment platforms like JobStreet in that it is designed specifically for the UiTM institutional environment. Everything from job posting to job screening and workflows is in a controlled and structured environment inside a university setting. The system can meet the increasing demand for an efficient, accessible and organised medium to support the internal recruitment process, while replacing the expensive manual or paper recruitment process often undertaken in an institutional environment.

This system is designed with a role-based system to clearly differentiate users, which consists of administrators representing the administration and HR personnel of UiTM and job seekers representing the persons who are seeking employment in the university. This separation allows the system to engage different user groups and even specific individuals according to their specific permissions. It ensures that they play their specific roles in the system, while guaranteeing that the data is kept safe, and it provides accountability for operating it during the recruitment process.

1.2 Purpose of the system

The goal of JobMatch is to digitalise and streamline the recruitment process of UiTM by replacing the manual and paper-based recruitment process with an integrated online recruitment process. The purpose of the system is to tackle the escalating demands for an efficient, accessible and systematic recruitment solution which allows employment management to be conducted easily across all UiTM campuses. From creating job announcements to publishing vacancies, accepting online applications, filtering, monitoring applications and hiring, JobMatch provides tools to manage the entire recruitment process. The automation of these processes will help to streamline recruitment and support a more transparent process, creating better experiences for persons applying for job positions within UiTM and also for the administrators.

2.0 GitHub Repository & YouTube Link

Github Repository Link: <https://github.com/eldamn1/jobmatch>

Link Presentation: https://youtu.be/OHuT_dPkNgU?si=cGKcy31Ze9j7bEzI



3.0 Entity-Relationship Diagram (ERD)

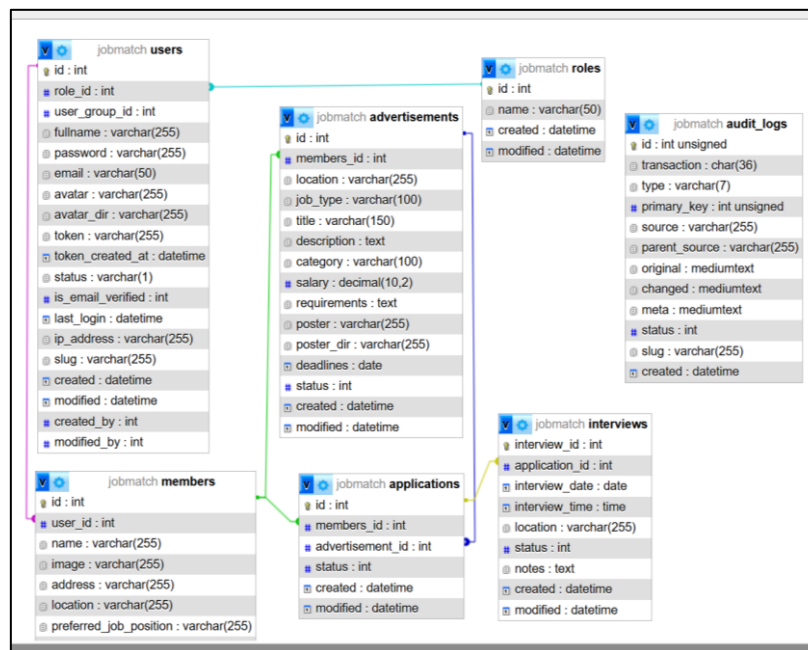


Figure 1: JobMatch's ERD

The Entity Relationship Diagram (ERD) presents a conceptual model of the JobMatch system component, the entities, their attributes and the relationships between them. This system consists of the following main components: Users, Members, Roles, Advertisements, Applications, Interviews, and Audit Logs. Each entity holds particular data which is needed for the efficient management of the system. For instance, the Users entity holds login details along with account information, whereas the Members entity contains extra personal information for the users (job seekers) as well as the admins. The access level for users is defined at the Role entity level, which allows for the appropriate levels of access for the administrators and users (job seekers) in the system.

The relations among these entities help realise the JobMatch system's core business processes. Each user gets assigned one role, which gives this user a profile with detailed personal information. Admins may publish multiple advertisements, and users (job seekers) may post multiple applications to various advertisements. The Interview entity is used with the recruitment workflow to schedule and manage interviews of applicants. It keeps interview information, including interview date, interview time and interview status, making the interview process well managed and monitored during the hiring procedure. The Audit Logs entity also logs any user activity, including login attempts and system activity, which can help you track users, bolster security, and help with system monitoring. Overall, the ERD serves as a blueprint for the desired data relationships, data integrity, and reduced data duplication, allowing for efficient and effective data management throughout the JobMatch system.

4.0 System Requirements

4.1 Software

The JobMatch system requires several software components to support the development, database management, testing, and execution of the web application. Each software plays an important role in ensuring that the system operates efficiently and that all features function correctly throughout the development and deployment process. The software requirements, including their versions and purposes, are presented in Table 4.1:

Software	Version	Primary Function
Laragon	V8.6.1	Local Development Environment: Provides Apache, PHP, and MySQL services to run the JobMatch system locally during development and testing.
PHP	PHP 8.3.30	Server-side scripting Language: Executes the application logic and process user requests in the JobMatch system.
MySQL	MySQL 8.4.3	Database Management System: Stores and manages application data such as users, job advertisements, applications, and members.
phpMyAdmin	V5.2.3	Database Administration Tool: Used to create, import, manage, and maintain the MySQL database through a web interface.
Visual Studio Code	1.128.0	Integrated Development Environment (IDE): Used to develop, edit, debug, and manage the project source code.
Google Chrome	Latest version	Web Browser: Used to access, test, and verify the functionality of the JobMatch web application.

Table 4.1: System Requirements

4.2 Version Compatibility

The JobMatch system requires compatible software versions to ensure that all components work together without compatibility issues. The selected versions of PHP, MySQL, Apache, CakePHP, and other development tools are compatible with each other and support the successful execution of the web application. By using compatible versions, the system can be developed, tested, and maintain in a stable local development environment. The version compatibility used in the JobMatch system is present in Table 4.2.

Component	Compatible Version	Compatibility Description
CakePHP	5.2	Compatible with PHP 8.1 or above and used as the main framework for developing the JobMatch system.
PHP	8.3.30	Fully compatible with CakePHP 5.2 and supports the required server-side functionalities.
MySQL	8.4.3	Compatible with CakePHP through PDO and used to store and manage application data.
Apache	2.4.66	Supports PHP 8.3.30 and is used to host the JobMatch application locally.
Laragon	8.6.1	Provides a compatible environment by integrating Apache, PHP, and MySQL for local development.
phpMyAdmin	5.2.3	Compatible with MySQL 8.4.3 for managing the project database through a web interface.
Google Chrome	Latest Version	Fully supports HTML5, CSS3, and JavaScript features for testing the web application.

Table 4.2: Version Compatibility

The software versions listed above are compatible with each other and help the JobMatch system run smoothly. The application is developed and tested using software that works well together. The user interface and system functions are also checked using a web browser to ensure that everything is displayed correctly. As a result, the JobMatch system can be developed, tested, and maintained in a stable and efficient development environment.

4.3 Dependencies

The Jobmatch application uses composer to manage its dependencies. The main packages required by the system are listed below.

Dependencies	Version	Primary Function
CakePHP	5.2	PHP Framework: Provides the MVC architecture, routing, controllers, models, and views for developing the JobMatch web application efficient.
Bootstrap	v5.x	Front-end framework: Used to create responsive, mobile-friendly, and modern user interface components.
Composer	2.10.2	Dependency Manager: Installs and manages CakePHP packages and third-party libraries required by the system.

Table 4.3: Dependencies

Overall, the software requirements and dependencies listed above are required to support the development and operation of the JobMatch system. Each software and dependency are used for a specific purpose during the development process. The source code is developed and edited using Visual Studio Code, while the database is created and managed using MySQL and phpMyAdmin. The application is executed locally using Laragon, and its functions are tested through a web browser. In addition, CakePHP, Bootstrap, HTML, CSS, and JavaScript are used to build the system structure, user interface, and system functionalities. By using these software tools and dependencies together, the JobMatch system can be developed, tested, maintained, and executed in a stable and efficient development environment. As a result, the overall system development process is supported effectively, and the application is able to perform its intended functions smoothly.

5.0 User Interface Overview

The JobMatch system features a modern, clean, and responsive user interface designed to provide an excellent user experience on both desktop and mobile devices.

5.1 Overall Layout Structure

The system uses a consistent layout throughout all pages with the following components:

5.1.1 SideBar

1. **System Logo:** Display at the official circular JobMatch UiTM logo at the very top.
2. **Main Navigation Links:** A clean list of active navigation links accompanied by visual icon:

(User Interface)

- Dashboard
- Profile
- Application

(Admin Interface)

- Dashboard
- Profile
- Advertisement
- Users
- Audit Logs

3. **Logout Button:** Positioned at the bottom of the sidebar.

5.1.2 Header

The Header is a clean, responsive top navigation bar that is composed by the slide toggle (hamburger icon) on the far left which expands/collapses the sidebar, and on the far right, it contains the crescent moon icon to switch them to dark and light version.

]

5.1.3 Main Content Area

In the main content area is dynamic user dashboard info that comes with a very neat card-and-grid arrangement. It is launched with a personalized greeting banner welcome user by Welcome back, Hawa (User). Immediately below is a search bar and filter bar that contains a text box that shows job titles, a dropdown that says All Status and a large golden-yellow search button.

Further down are four metric cards to give a quick glance at application data, that has shown that the user has applied a total of 1, there is 1 pending, 0 approved, and 0 rejected.

After these statistics are two quick access buttons which are Apply for a Job and a white coloured button that simply reads Update Profile and has a clean dark border. Below is a two-column grid that has recent applications table (job position, date and status, a yellow coloured pending badge) in the first column while a section highlighting the latest available job openings in the other column, such as Barista at Cafe Jasmine, UiTM Puncak Perdana, and an Apply button for that job opening.

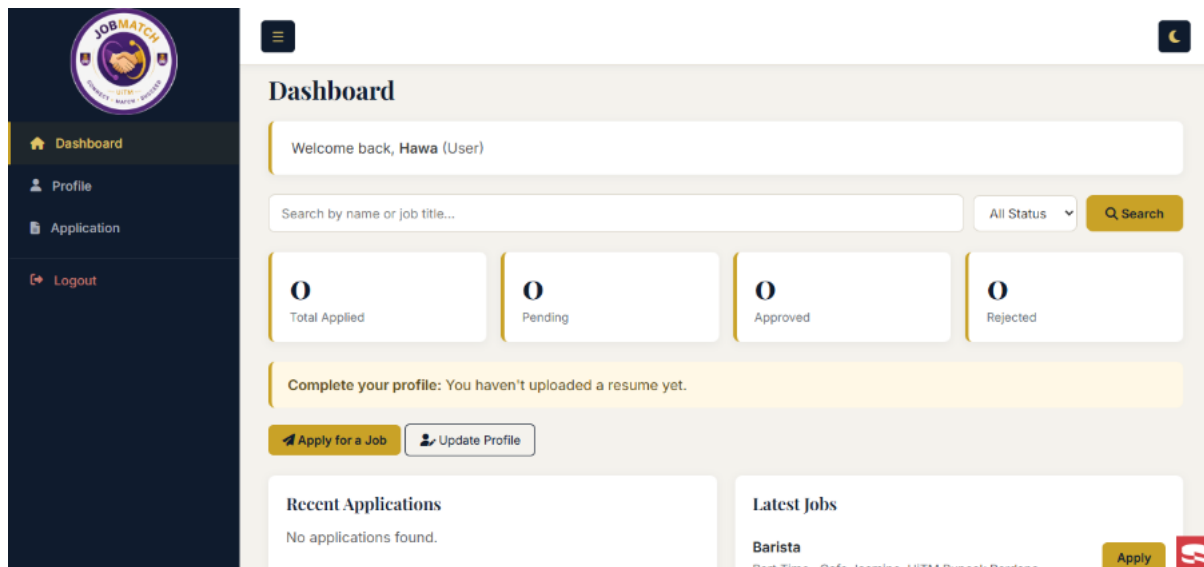


Figure 2: Layout JobMatch

5.2 Navigation System

The navigation system adapts based on the user's role:

Applicant Navigation Menu:	Administrator Navigation Menu:
Dashboard	Dashboard
Profile	Profile
Application	Advertisement
Logout	Users
	Audit Logs
	Logout

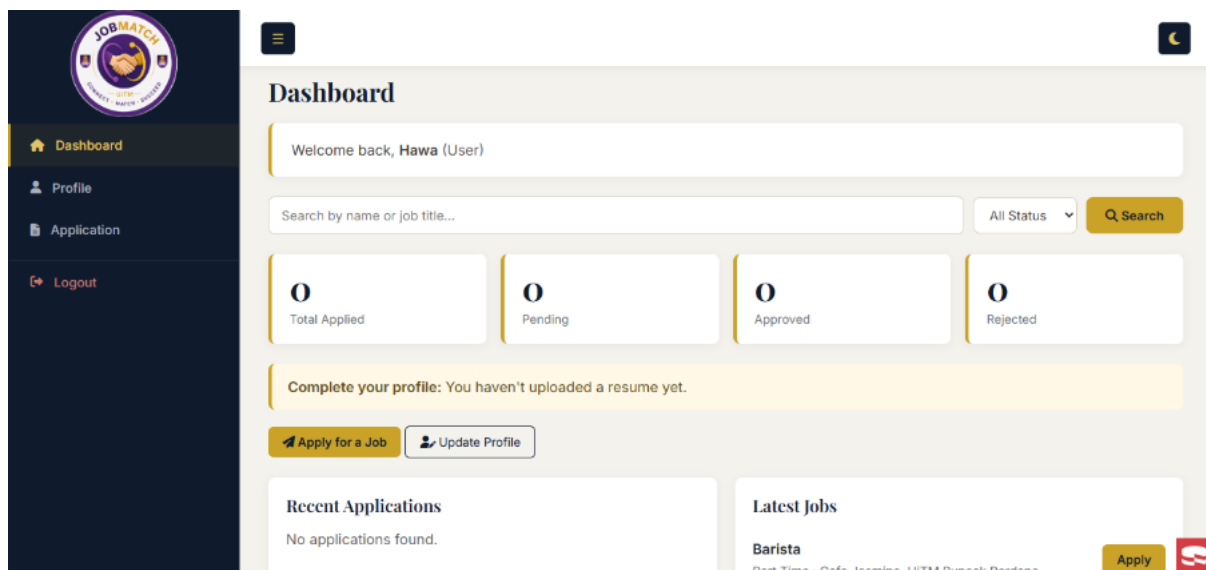


Figure 3: Navigation bar for applicant

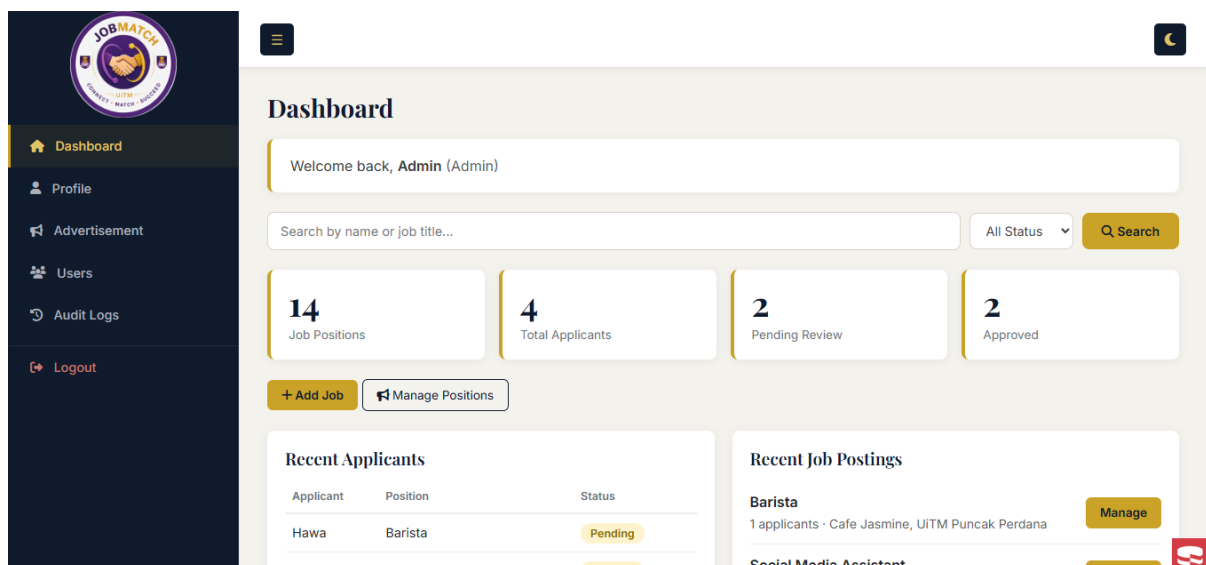


Figure 4: Navigation bar for admin

6.0 Features & Functionalities

1. Interactive Job Advertisement

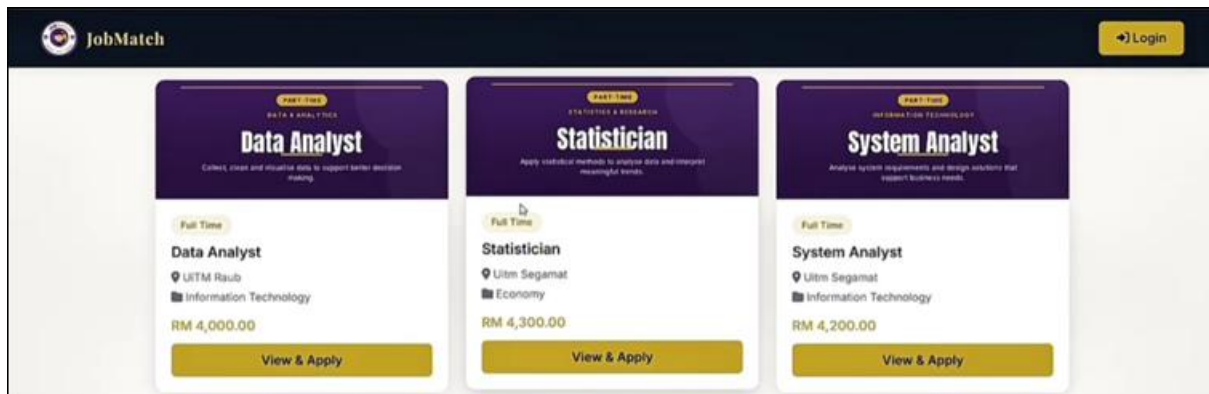


Figure 5: JobMatch Advertisement Page

The Interactive Job Advertisement function acts as the primary access point of the JobMatch website, where users are able to review any available positions on the site before registering to the website. The advertisements have a brief description of the job summary, which includes the name of the job, the company that offers the job, and its location. Once users click on the desired job advertisement, they are able to view the job description, which highlights the job responsibilities, the qualifications that required, the salary details (if available), and the deadline date for the job.

2. Smart Job Search

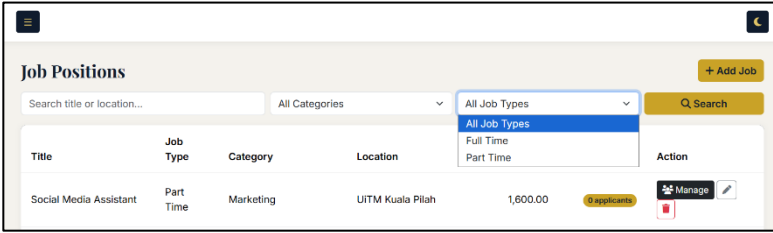


Figure 7: Job Search 1

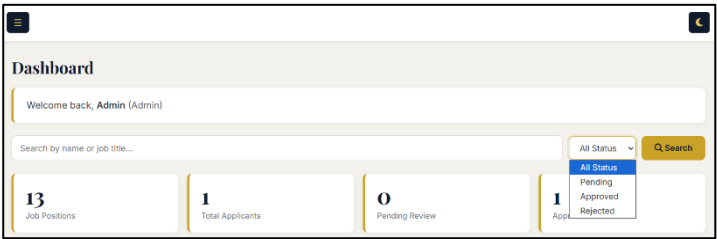


Figure 6: Job Search 2

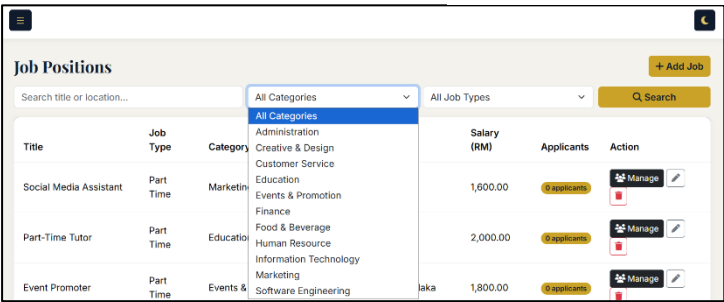


Figure 8: Job Search 3

Smart Job Search features is aimed in helping users to find appropriate jobs effectively without searching through all job vacancies available one by one. Users may look for jobs by entering keywords such as job title, company name, or any necessary skills. Thus, users will be able to get their desired jobs in faster way. All search results will appear instantly, enabling users to make comparisons between various job vacancies instantly with their preferences. This functionality considerably enhances the entire user experience, making job finding faster and more comfortable for users.

3. Resume Upload

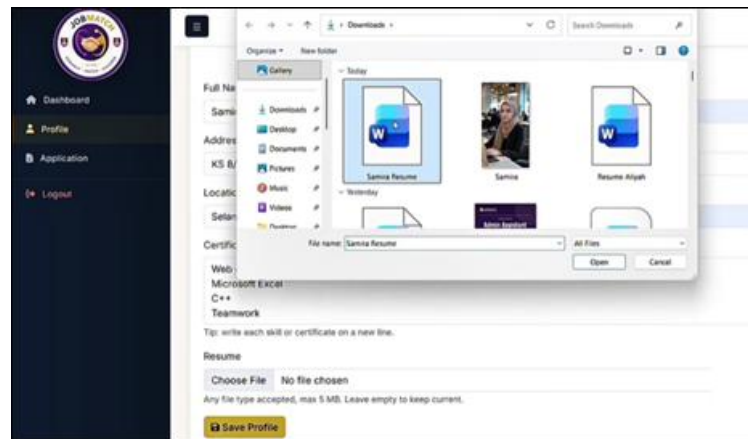


Figure 9: Resume Upload

Resume Upload features make the application process easier by helping people upload and maintain their resumes using the profile page itself. Once the resumes have been uploaded, the person does not need to provide all the same information every time they want to apply for different job vacancies. It is a quick way to make applications, and at the same time, makes sure that employers get the latest version of the resume of an applicant. In reducing repeated tasks, the system promotes user to apply for more options and makes the entire recruitment process more efficient.

4. Admin Recruitment Dashboard

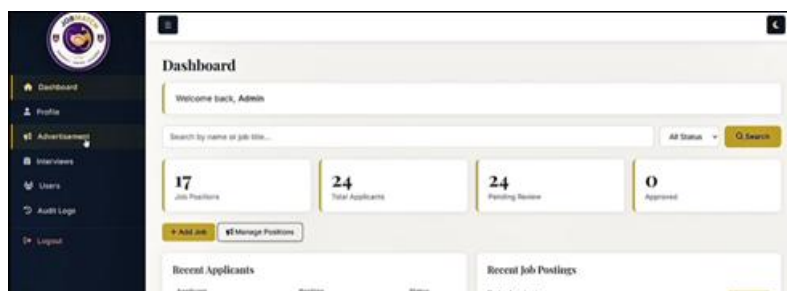


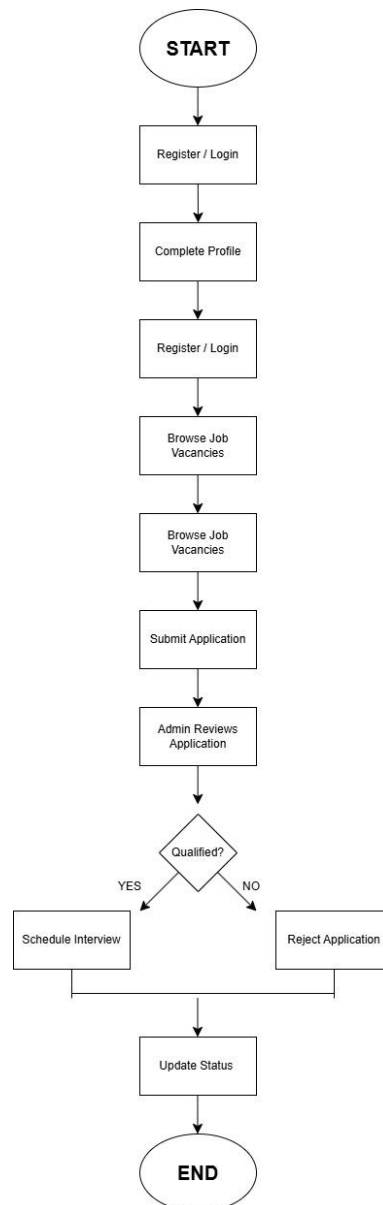
Figure 10:Admin Dashboard

The Admin Recruitment Dashboard gives a detailed insight into all the recruitment processes conducted within the JobMatch website. This tool presents as vital information that show the total amount of received job applications, vacant job posts, applicants' activity, and system statistics. At the same time, administrators can search for particular records, create new job advertisement posts, and control all vacant jobs from one screen. In this way, the administration can control the hiring process and make quick decisions, when necessary, through the presentation of recruitment information on a dashboard.

7.0 Workflow of Form

Below is the breakdown of the step-by-step workflow in the job matching system from the beginning of the process (user registration) to the end of the system process (job application result): It involves the cooperation between the user (applicant) and the system administrator.

Here is the detailed breakdown of each step in the workflow:



8.0 Team Roles and Responsibilities

No.	Name	Roles	Responsibilities
1	Aisyah Maisarah binti Seliman	Project Leader (Lead System Developer)	Developed the JobMatch application with a full-stack approach using CakePHP 5 Framework. Developed the authentication system with login, session and logout functionality. Created role-based access control (RBAC) for the jobs system to distinguish between the user anonymity of Admin and Job Seeker. Developed an advertisement module which involves the creation of job postings, editing, deletion, and management of applicants, including the ability to approve and reject. Improved the dashboard module for both Admin and Job Seeker views, adding statistic summary cards, recent job listings and role-based content rendering
2	Nik Nur Aliyah Safiah binti Nik Nazree	Database Designer & UI/UX Lead	Developed and designed the Entity Relationship Diagram (ERD) - Identified entities, defined attributes, cardinality mapping and normalised to Third Normal Form (3NF). Defined primary keys, foreign keys and table relationships across the database schema. Strategically overseeing the visual appearance, aesthetic theme, layout consistency and overall user interface and user behaviour of the UI system, such as navigation, typography and colours.
3	Nurul Syazwina binti Shoib	Backend Developer	Created the Users Management module to help Administration users manage, monitor and view all users registered in the system. Installed the Audit Logs module to monitor and log all changes to data such as creates, updates and deletes, to give administrators a history of what has happened within the system to facilitate data governance and accountability.

4	Hawa binti Mohd Zamri	Module Developer	Handled the end-to-end application job submission workflow and designed and implemented the Application module. Design of application listing screen. Handled functionality for application status updates for approved, pending and rejected applications by the admins.
5	Farah Syazana binti Mat Rohani	Module Developer	Handled the design and implementation of the Profile module, which allowed job seekers to enter their personal information such as full name, contact information and address. Designed and implemented resume upload functionality, which supported the PDF file format and managed file handling and storage on the server side.

9.0 Contact Information

The following project roles and responsibilities have been assigned to the team members to ensure that the JobMatch system is developed, maintained, and supported effectively. Each member is responsible for a specific module or system component, contributing to the successful implementation and operation of the web application.

Project Role	Responsibility	Contact Details
Project Leader (Lead System Developer)	Managed the development of the JobMatch system using the CakePHP 5 Framework. Created the login, logout, and session functions. Set up different access for Admin and Job Seeker users. Developed the Advertisement module to add, edit, delete, and manage job postings and applications. Improved the dashboard by adding summary information and different content based on user roles.	Name: Aisyah Maisarah binti Seliman Email: 2025219402@student.uitm.edu.my
Database Designer & UI/UX Lead	Designed and developed the Entity Relationship Diagram (ERD), including database tables, relationships, and keys. Managed the user interface (UI) design by ensuring a consistent layout, simple navigation, and a user-friendly experience.	Name: Nik Nur Aliyah Safiah Binti Nik Nazree Email: 2025248724@student.uitm.edu.my
Backend Developer	Developed the User Management module to help administrators manage and view registered users.	Name: Nurul Syazwina Binti Shoib Email: 2025235018@student.uitm.edu.my

	Implemented the Audit Logs module to record user activities such as creating, updating, and deleting data for easier monitoring and tracking.	
Module Developer	Developed the Application module to handle the complete job application process. Designed the application listing page and implemented status management, allowing administrators to approve, reject, or keep job applications as pending.	Name: Hawa Binti Mohd Zamri Email: 2025483672@student.uitm.edu.my
Module Developer	Designed and developed the Profile module to allow job seekers to update their personal information, including full name, contact details, and address. Implemented the resume upload feature with PDF support and managed file uploads and server storage.	Name: Farah Syazana Binti Mat Rohani Email: 2025237764@student.uitm.edu.my

Table 10.1 Contact Information (Support)

10.0 Conclusion and Reflection

The creation of JobMatch website proves to be a successful implementation of concepts and principles of web development and system design that have been acquired. The objective of the project was to create a system which could help in connecting employers with job seekers using an easy, efficient, and effective process of recruitment. Using various essential features, JobMatch has fulfilled the fundamental requirements of an online recruitment website. During the process of development, some technical aspects were also integrated in the system, such as database design with the help of ERD (Entity-Relationship Diagram), user interface design, form workflow, and CRUD (Create, Read, Update, Delete) operations.

This project has been a very educational journey for all of us team members as we have learned a lot about the entire process of system development, which included planning and designing the database to its implementation and testing. We gained knowledge on web development technologies, database management, system integration, and user-oriented design while fulfilling the functional and technical requirements for the website. Moreover, by working in a team, we were able to enhance our skills of communication, collaboration, time management, and problem solving as all of us had a different role in this project. We have faced various issues during this project, including debugging system problems and module integration but they have made us learn more about critical thinking and problem solving.

In general, the JobMatch website has been successful in meeting its goal of creating a working and well-designed website recruiting system that works to benefit the both job seekers and administrators. Apart from meeting the criteria of the assignment, we have learned how to create a web application through the process of creating this system. This project has emphasized the importance of proper planning, good teamwork, constant testing, and user-centered design while creating an information system. We believe that the experience gained in this project will be useful to our future academic and professional endeavours in information systems management.